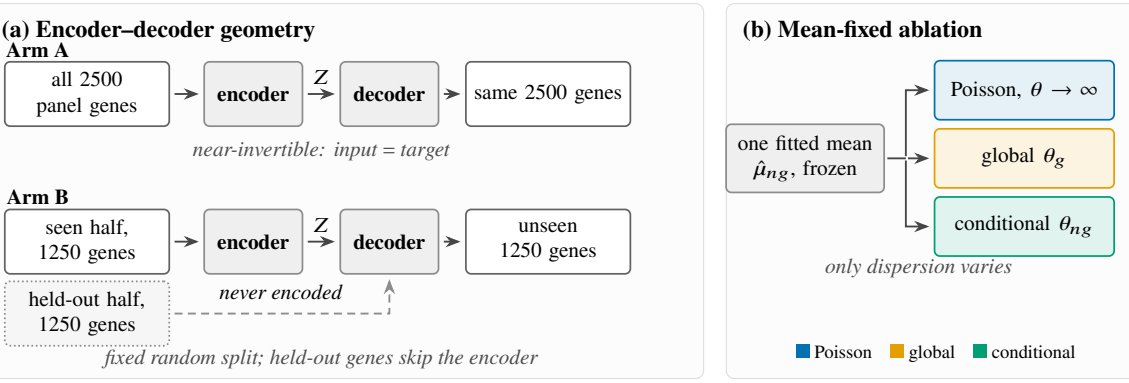

(a) Encoder–decoder geometry
Arm A
all 2500 panel genes
encoder
Z
decoder
same 2500 genes
near-invertible: input = target
Arm B
seen half, 1250 genes
encoder
Z
decoder
unseen 1250 genes
held-out half, 1250 genes
never encoded
fixed random split; held-out genes skip the encoder
(b) Mean-fixed ablation
one fitted mean $\hat{\mu}_{ng}$, frozen
Poisson, $\theta \to \infty$
global θ_g
conditional θ_{ng}
only dispersion varies
Poisson
global
conditional